

# ASHENFALL — Build Phases (M0–M9)

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**Companion document to DESIGN.md (v0.7)** · Every milestone, what it adds, what it must pass before the next one starts Prepared 2026-09-26

This is the checklist version of DESIGN.md §14 — expanded so that each phase has an **entry condition**, a **deliverable list**, a **verification gate**, and an **explicit "do not proceed until"** rule. Nothing gets built until you approve the design; nothing gets committed until its gate passes.

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## How to read this

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| Field     | Meaning                                                                                                                             |
|-----------|-------------------------------------------------------------------------------------------------------------------------------------|
| Adds      | The mods/content that enter the pack in this phase. Anything not listed here is <i>not</i> in the pack yet — no exceptions.         |
| Gate      | The test that must pass before the next phase begins. If a gate fails, we fix it <i>in that phase</i> , we do not carry it forward. |
| Lock      | Something that, once set, must never be changed without a new world or a full re-test. Worldgen and water are the big ones.         |
| Decisions | The D-numbers from DESIGN.md §15 that this phase forces us to settle.                                                               |

**The one rule that matters most:** *one phase at a time, and no mod enters the pack without being on its phase's Add list.* Phases exist so

that when something breaks, we know exactly which of ~130 mods did it.

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## Phase map

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|     |                        |       |                                 |
|-----|------------------------|-------|---------------------------------|
| M0  | Skeleton               | ————— | can it boot at 60 FPS with noth |
| M1  | Movement & combat      | ——    | does the vanilla-combat layer f |
| M2  | Soulslike layer        | ———   | does death mean something?      |
| M3  | Worldgen               | ————— | does the world have the right s |
| M3b | World fullness & water | —     | does the world feel lived-in    |
| M3c | Dungeon layer          | ————— | is there something to find in e |
| M4  | Nations                | ————— | who do I owe, and who hates me? |
| M5  | RPG & grinding         | ———   | why do I kill this mob a hundre |
| M6  | Boss spine             | ————— | the difficulty staircase        |
| M7  | Landmarks & story      | ——    | the actual plot                 |
| M8  | Crafting & loot        | ———   | the gear ladder                 |
| M9  | Balance & perf         | ———   | the full playthrough pass       |

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## M0 — Skeleton

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**Effort:** 1 session · **Lock:** nothing yet · **Decisions:** D12, D13

### Adds

- Packwiz project (`pack.toml`, `index.toml`) + PrismLauncher/ CurseForge-compatible profile
- `overrides/` `skeleton: config/, kubejs/, resourcepacks/, shaderpacks/` (*empty*), `patchouli_books/`
- Perf stack only: **Embeddium + Extra/++**, **ModernFix**, **FerriteCore**, **ImmediatelyFast**, **Entity Culling**, **Clumps**, **Alternate Current**, **Enhanced Block Entities**, **FastSuite**, **MoreCulling**, **Structure Layout Optimizer**, **Dynamic FPS**
- **Acedium Sodiumized** (Nvidia) with the `compat` profile as fallback

- **Chunky**
- Mod budget statement written into `README` in the pack folder:  
≤130 content + ≤18 utility/performance

## Gate

1. Boots clean, no crash, JEI opens.
2. 60+ FPS at render distance 10–12 in a vanilla-ish test world.
3. ImmediatelyFast's `experimental_screen_batching` tested against a GUI (JEI, and any backpack GUI once M3b lands) — either proven safe or disabled.
4. Acedium activates on your GPU; the `compat` profile falls back to stock Embeddium without error.
5. RAM: 6 GB allocation, no GC storms; 8 GB only if measured.

**Do not proceed until:** the perf baseline is *measured and written down*. Every later phase gets compared against it.

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## M1 — Movement & combat

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**Effort:** 1 session · **Decisions:** D1 (settled)

### Adds

- **Backported Spears** (jab, speed-scaled charge, Lunge, Mob Kabob) — with its configs set: `lunge_hunger_cutoff`, `can_use_lunge_while_gliding`, `can_use_lunge_while_in_water`
- **ParCool + ParCool+ Compatibility++** + Player Animator (via the addon)
- **Combat Roll** — bound to a dedicated key; ParCool's fall-roll disabled or the double-trigger resolved
- **GhostSwap** (our custom NeoForge mod, **source already written** in `ghostswap/`) — attribute swapping on key G, save on H/J; build with `./gradlew build`, rename the jar to `ghostswap-1.21.1-neoforge-0.1.0.jar`
- **Simply Swords, Spartan Weaponry (unofficial)** — with

Spartan's spears disabled (Backported Spears owns that slot)

- Config pass on **Backported Spears vs Simply Swords vs Spartan** weapon overlap

## Gate

1. Rolls, vaults, wall-runs, ledge grabs all work with no pose breakage.
2. Rolls do not double-trigger with ParCool's fall-roll.
3. Spear charge damage scales with movement speed; Lunge works on foot, on a boat, on a horse, while falling.
4. GhostSwap sword→mace→sword swaps damage/attributes **within one tick** — this is the MC-28289 check, and it is the one thing in the mod that must be verified in-game (see `ghostswap/README.md` for the fallback if it lags).
5. Spear Lunge does not glitch through terrain.

**Design decision to make here:** whether **rolling cancels a spell cast** (see M3c) — the interaction is set now so it isn't retuned twice.

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## M2 — Soulslike layer

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**Effort:** 1 session · **Decisions:** D4 (settled at -20%)

### Adds

- **Silent's Power Scale** (*replaces Scaling Health, which has no 1.21.1 build — same author, generalises it to any attribute on players and mobs*)
- **GraveStone Mod**
- (*Soulslike Deaths dropped — Modrinth has 1.21.11 and 1.21 Fabric only, no 1.21.1 NeoForge build*)
- Custom **Ember Flask** (KubeJS) — no natural regen, refills only at rest points
- Death ruleset: Embers dropped at death, one recovery run, Hollowing capped -20%

## Gate

1. Natural regen is off; the Ember Flask is the only heal.
  2. On death, Embers drop at the death location and exactly one recovery run is possible.
  3. Hollowing never exceeds -20%, and Hollowing does not affect anything else.
  4. GraveStone is a locator only — it does **not** protect inventory.
  5. Distance scaling at 1k / 5k / 15k blocks reads as "hard", not "absurd", with no double-scaling spike.
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## M3 — Worldgen (the world's shape)

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**Effort:** 2 sessions · **Lock:** after this phase, the world is generated.

**Never add or remove a worldgen mod without a new world.** ·

**Decisions:** D6, D8

### Adds

- **Terralith** (+ Terrablender) — Tectonic **off by default**
- **Explorify, Dungeons and Taverns, Towns and Towers, When Dungeons Arise**
- Density tuning across all of them, then **Chunky starter-region pregen** ~10k × 10k + on-demand pregen profile

## Gate

1. All five biomes-of-interest generate and appear in the Heartland within reasonable travel distance.
2. Structure density check: the world is not flooded; a 10-minute walk finds at least one point of interest.
3. Explorer's Compass blacklist is set (noise structures removed) so searches stay fast.
4. Chunky pregen completes without freezes; save size is acceptable.
5. Tectonic verified *off* in the shipped config, with the optional profile documented.

**Do not proceed until:** the water stack decision (M3b) is *scheduled next*, because Dynamic Waters requires a brand-new world — if it comes after this, we regenerate.

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## M3b — World fullness & water

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**Effort:** 2 sessions · **Lock:** the water stack must be locked before this world exists. If M3 already generated the world with water mods absent, regenerate.

### Adds

- **FallingTree** — logs-per-cut gated by Woodcutting level; shift-click bypass
- **Sophisticated Backpacks** (+ Core) — tool swapper, feeding, filters, tank, mob catcher, void, placeable
- **SWEM** — horse gaits, training, bonding, genetics, tack, nation horse lines
- **Guard Villagers** — raid defence becomes the standing system's social lever
- **Farmer's Delight** — cooking, and the food tier the Ember-Warden cares about
- **Dynamic Waters + Flowing Fluids + Boatload** (*Streams Reflowing is the alternative to Dynamic Waters — never both*)
- **Sable + Sablewaves** (*optional, perf-gated*), **Currents of Trade** (*optional*)
- **Supplementaries, Comforts, Tool Belt** (*optional*)

### Gate

1. Rivers flow downhill; boats are pushed by streams; ocean currents exist.
2. FallingTree respects the Woodcutting cap; giant Terralith trees don't trivialise wood.
3. Backpack Curios integration tested — disabled if it crashes.
4. Guard spawns don't overrun Millénaire villages.
5. Water physics cost measured: tick time stays stable in oceans and

dams; apply the §6.11 cut order if not.

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## M3c — Dungeon layer

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**Effort:** 2 sessions · **Decisions:** D5, D15

### Adds

- **YUNG's suite (×9)** — Better Dungeons, Strongholds, Mineshafts, Desert Temples, Jungle Temples, Witch Huts, Ocean Monuments, Nether Fortresses, End Island
- **Repurposed Structures** — every variant explained in-world as a *displacement scar*
- **Hopo Better Ruined Portals / Underwater Ruins** (*Hopo's mineshaft is excluded — YUNG's owns mineshafts*)
- **The Graveyard** — the Lich's Bargain chain, with its villager-blood requirement reworked in KubeJS
- **Structory, Explorify** density pass
- **~~Ancient Warfare 2~~** — **dropped, no 1.21.1 build**. Its role as the Empire's fallen provinces moves to Towns and Towers + Structory + Repurposed Structures, with a KubeJS province register in `structures.js` naming each generated structure as a ruin of a specific fallen province
- **Dimensional Dungeons** — the renewable dungeon dimension for the grinding layer and the 12 Hollow Challenges
- **Iron's Spells 'n Spellbooks + Monsters & Spellbooks** — Catacombs, Lost Citadel, Echo of Tyros, and the magical structures holding runestones
- **Lootr** on every structure chest; loot-table injection for Codex pages, unique loot, runes and manuscripts

### Gate

1. Every vanilla structure type is replaced and renamed; nothing generates "as vanilla".
2. Structure density and spawn rates tuned; no mineshaft duplication (YUNG's only).

3. Lootr works on every structure mod's chests — no duplicate or empty chests.
4. The **Iron's Spells chain** works end-to-end: Scroll Forge → Inscription Table → Arcane Anvil → weapon imbuing; wizard scroll trades stripped in KubeJS.
5. The **Necro school** loads without destabilising the base mod (drop the addon, never the base mod).
6. The nine Ember Runes are all findable and none is soft-locked.
7. Eldritch Manuscripts are obtainable only from the intended late-game sources.

**Do not proceed until:** the Dungeon Layer's density is *final*. Every structure added later changes every compass search and every pregen estimate.

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## M4 — Nations

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**Effort:** 1–2 sessions · **Decisions:** D7

### Adds

- **Millénaire** (7 cultures) — *or* the fallback: vanilla villages + Towns and Towers + KubeJS-only factions
- ~~**Ancient Warfare 2**~~ — out. Nation *identities* come from Millénaire's seven cultures plus the KubeJS standing system; nation *set-pieces* come from Towns and Towers / Structory / Repurposed Structures
- The **standing system** in KubeJS + FTB Quests: –100...+100, nine nations, cross-faction consequences
- Nation gear lines, tribute, champion quests, the nine nation ultimates
- Hexer's –40 with the Witchbane Crusaders; class-based standing penalties
- **Narrative:** Aviel's Dialogue Mod (ADM) — the dialogue engine, JSON dialogues + NPC templates shipped as a datapack; first ADM dialogues for the nine nations; the culture material shops

**Gate 0.** `/npc validate` passes on every dialogue file shipped so far

1. Villages generate, grow, and don't erase each other.
  2. Standing changes measurably: +10 for defending a village, -10 for raiding one.
  3. Cross-faction penalties fire (helping one nation's enemy costs standing with them).
  4. You **cannot** be champion of all nine — verified by attempting it.
  5. Millénaire boot test passes, or the fallback is switched on with the decision recorded.
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## M5 — RPG & grinding

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**Effort:** 1 session · **Decisions:** D9

### Adds

- **Project MMO + Classes + Skill Books + XP Bottles + Origins / Origins: Classes**
- **Mob Grinding Utils** — fans, absorption hoppers, masher
- **Reforged** — craftable tiers/modifiers + anvil stat rerolling
- **Attribute Modify** — datapack attribute tuning (base attributes only; must not touch Reforged's dynamic NBT or the `irons_spellbooks` namespace)
- **Mob Champions + The Ashenfall Threat** tiers (seven per region)
- **Loot Beams: Refork** with rarity colours mapped to Reforged's tiers
- Expert recipe rebalance: no villager power, netherite needs a wither skull, modded recipes gated by skill level

### Gate

1. Project MMO + Reforged + Attribute Modify all apply together; base vs dynamic attribute ownership is clean.
2. Skill levels give real stats; skill books drop from bosses and dungeons.
3. Champions stay capped under distance scaling; Tier I Heartland

- never out-threats Tier VII Frontier.
4. Loot beam colours match Reforged tiers.
  5. The magic attribute stack (18+ attributes) doesn't collide with the melee attribute stack.
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## M6 — Boss spine

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**Effort:** 2 sessions · **Decisions:** D5

### Adds

- **Cataclysm** + Lionfish API + Curios (Ignis, Netherite Monstrosity, Ender Golem, Leviathan, Ancient Remnant, The Harbinger, Scylla, Maledictus)
- **BOMD** (Night Lich, Obsidilith, Nether Gauntlet)
- **The Twilight Forest** (Naga, Lich, Hydra, Ur-Ghast, Final Boss)
- **The Aether** (Slider, Valkyrie Queen, Sun Spirit)
- **Deeper and Darker** (Otherside)
- **Alex's Caves (unofficial port)** — Chapter 6 made skippable
- **Mini-boss Boss Bars, Configurable Boss Bars, Boss Music Mod, Floating Damage Indicators, YUNG's Traveler's Titles**
- The 21-boss order test and the 12 Hollow Challenge arenas

### Gate

1. Every portal works; no dimension ID collisions.
  2. Every boss is killable with vanilla mechanics — no boss requires a modded move-set we don't have.
  3. Boss order matches the tier ladder: no boss is beatable with gear from two tiers down.
  4. Boss bars are consistent across all 21 bosses + trials.
  5. Alex's Caves boots without crashes; Chapter 6 can be skipped cleanly.
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## M7 — Landmarks, puzzles & story

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**Effort:** 3 sessions · **Decisions:** D2, D10, D11

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## Adds

- The **Landmark Chain** (L1–L20 + L★) as actual structures and coordinates
- The four discovery systems: **Explorer's Compass** (blacklist-gated), **Cartographer's Maps** (dungeon loot), **Rumours** (FTB Quests + KubeJS dialogue), **The Pilgrim's Journal** (treasure-map pages)
- **The Empire's Annals** (12 texts, in order, the ending gate)
- **The Pilgrim's Journal** (24 pages) + the **Lore Codex** (24 entries)
- All puzzle types as datapack structures + KubeJS: Riddle Vaults, Elemental Gates, Pedestal Ordering, The Labyrinth, Parkour Trials, 12 Combat Trials, Lever/Song Puzzles
- **The nine Ember Runes** and the **Hidden Cathedral**
- **The full narrative pass:** ~30 ADM dialogue files, ~20 named NPCs, ~15 rumours, all 21 boss bar names, 6 full monologues, the Patchouli books, and the resource packs (portraits, dialogue UI, item textures, boss audio)
- **FTB Quests chapters 0–8** with the real narrative text from LORE.md
- Patchouli books, waystones per chapter, the NG+ hook

## Gate

1. **No gate is soft-locked.** Every landmark is reachable by at least two independent routes; every rune is findable.
  2. The Annals must be found in order, and all 12 are required to locate the Cathedral of Ash.
  3. All 24 journal pages exist and are obtainable.
  4. All 24 Codex entries are written and injected into their structures.
  5. All nine runes open the Hidden Cathedral, and the cathedral cannot be opened without them.
  6. Chapters 0–8 are playable start to finish with no dead-end quest.
  7. `/npc validate` clean; no placeholder text in any dialogue, journal page or Codex entry
  8. Every rumour points at a landmark reachable by a second route
  9. **Act I (Ch 0–5) is shippable on its own** — D2's deliverable.
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## M8 — Crafting & loot

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**Effort:** 2 sessions

### Adds

- The full crafting chain: vanilla smithing → Mechanical Fusion Anvil → Aether Altar/Freezer → Scroll Forge / Inscription Table / Arcane Anvil → Hollow Court infusion → **Ashen Anvil** (KubeJS)
- **Signature weapons** — one per boss, each with a gimmick and an **imbued-spell slot**
- The **nine nation ultimates** (+100 standing each)
- KubeJS recipe gating: skill levels, boss-material flags, expert rebalance
- The two trade shops (Guild of Merchants, culture materials) — **materials only, audited**
- The **Ashen tier** (from the Hidden Cathedral) and its core material
- Reforged hammer currency, repair economy, Guild of Merchants pricing

### Gate

1. Every tier's gear is craftable and every gate is checkable.
2. No vendor sells power — verified by auditing every trade table **and every ADM trade file**
3. Every signature weapon upgrades through the ladder using the next boss's material.
4. Imbued weapons work for melee builds that never cast a spell.
5. The Ashen tier's material has exactly one source (the Hidden Cathedral).

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## M9 — Balance & perf

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**Effort:** 2 sessions · **Decisions:** D12, D13 (final verification)

## Adds

- Full playthrough pass, start to finish, with logging
- FPS/RAM verification on your GPU (3500-series class)
- Entity caps, particle settings, and the §6.11 / §13 **cut order** applied if any budget is missed
- `performance` / `compat` / `magic` profile variants finalised
- The mod-budget audit: does the shipped pack stay  $\leq 130$  content +  $\leq 18$  utility?
- Port-risk register re-check (Acedium, Alex's Caves, Enigmatic Legacy+, Millénaire forks)

## Gate

1. 60+ FPS at render distance 10–12 with no shaders, in a boss arena, in a flooded structure, and in the End.
2. Tick time stable with a large water volume and a mob farm running.
3. Full playthrough completable with no crash, no soft-lock, no missing lore.
4. All three/four endings reachable.
5. The cut order is written into the pack README so it can be applied by anyone.

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## Cross-phase registers — the things that must not be forgotten

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| Register                | Count | Where it lives      | Phase |
|-------------------------|-------|---------------------|-------|
| Bosses in the gauntlet  | 21    | DESIGN §10          | M6    |
| Nation champions        | 9     | DESIGN §10, LORE §7 | M4    |
| Hollow Challenge trials | 12    | DESIGN §8           | M7    |

|                              |             |                                   |          |
|------------------------------|-------------|-----------------------------------|----------|
| The Empire's Annals          | 12          | LORE §9                           | M7       |
| Pilgrim's Journal pages      | 24          | LORE §10                          | M7       |
| Lore Codex entries           | 24          | DESIGN §5.5                       | M3c + M7 |
| Ember Runes                  | 9           | DESIGN §8.3, LORE §13             | M3c + M7 |
| Landmarks                    | L1–L20 + L★ | DESIGN §5.3                       | M7       |
| Threat tiers per region      | 7           | DESIGN §6.4                       | M5       |
| Skill tracks                 | 8 + magic   | PROGRESSION §4                    | M5       |
| Spell schools                | 9 + Necro   | DESIGN §6.6                       | M3c      |
| Signature weapons            | ~12         | DESIGN §11                        | M8       |
| Nation ultimates             | 9           | DESIGN §11                        | M8       |
| Endings                      | 4           | LORE §13                          | M7       |
| Structure types reworked     | 15 (Tier A) | DESIGN §5.5                       | M3c      |
| Music tracks / discs         | ~10         | LORE §12                          | M6       |
| Named NPCs with dialogue     | ~20         | NARRATIVE-SYSTEMS §1              | M4 + M7  |
| Dialogue files               | ~30         | NARRATIVE-SYSTEMS §1              | M7       |
| Rumours                      | ~15         | DESIGN §5.2, NARRATIVE-SYSTEMS §2 | M7       |
| Trade shops (materials only) | 2 files     | NARRATIVE-SYSTEMS §9              | M8       |

|                              |                          |                      |         |
|------------------------------|--------------------------|----------------------|---------|
| Boss monologues              | 21 (one line) + 6 (full) | NARRATIVE-SYSTEMS §5 | M6 + M7 |
| Verification checklist steps | 22                       | MODS-INTERACTIONS §8 | all     |

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## What "done" looks like

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The pack is finished when:

1. M9's gate passes with numbers written down.
  2. Every register above is complete — no placeholder entries.
  3. The mod budget is within  $\leq 130$  content +  $\leq 18$  utility.
  4. Act I is playable end-to-end with no dead ends, and the ending gate is reachable.
  5. The cut order, the port-risk register and the update protocol are all in the pack folder.
  6. All five design documents still describe what actually shipped.
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*Phases, gates and locks are specified in DESIGN.md §14. Verification steps per mod are in MODS-INTERACTIONS.md §8.*